# Calibrating Temperature Sensors

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## Introduction and Setup

In this experiment, we calibrated an NTC thermistor using a **DS18B20** digital thermometer as the reference. The DS18B20 works from −55 to +125 °C, is accurate to ±0.5 °C, and has a 9–12 bit resolution.

The circuit is a simple voltage divider with the NTC connected on one side to GND. (We tried both configurations and got approximately similar readings within a 99% confidence interval.) A 10-bit ADC (0–1023 counts) reads the divider. For the calibration we collected 3 runs: one heating and two cooling. After cleaning, we had 1078 heating and 4243 cooling samples from 0 to 98 °C. We fitted the readings in two ways: polynomials of degree 1 to 3, and the physical NTC β-model. We also measured how fast the thermistor responds (its time constant).

## Raw Data

Temperature falls steadily as ADC counts rise, but not linearly: the curve is steeper at low counts. The time plots show a slow, smooth change in bath temperature.

## Heating and Cooling Calibration

**Table 1.** Data audit before fitting.

| Item | Heating | Cooling |
|---|---|---|
| Samples logged | 1079 | 4252 |
| Empty (NaN) rows | 1 (ADC 849) | 2 |
| Outliers removed ($> 3\sigma$) | 0 | 7 (> 1.66 °C) |
| Samples used in fit | 1078 | 4243 |

We fitted polynomials of degree 1 to 3 to temperature against ADC. Empty rows and outliers (beyond 3 standard deviations of the first cubic fit, or 1.66 °C for the cooling runs) were removed first. We used the common 3σ cut; the earlier 2.5σ cut removed 2 more points and changed the RMSE by less than 0.01 °C.

**Table 2.** Fit errors (°C) for heating (H, $N = 1078$) and cooling (C, $N = 4243$).

| Deg. | $R^2$ (H) | RMSE (H) | Max (H) | $R^2$ (C) | RMSE (C) | Max (C) |
|---|---|---|---|---|---|---|
| 1 | 0.99600 | 2.326 | 3.913 | 0.99265 | 1.462 | 4.010 |
| 2 | 0.99718 | 1.951 | 4.146 | 0.99882 | 0.586 | 2.752 |
| 3 | **0.99991** | **0.345** | **1.459** | **0.99898** | **0.546** | **1.727** |

The cubic fit is the best. The straight line leaves an S-shaped error of up to about 4 °C. The cubic cuts the RMSE to 0.35 °C for heating and 0.55 °C for cooling. All cubic terms are significant ($|t| > 17$, $p < 0.001$). The large condition number ($\sim 10^9$) comes from using raw ADC powers and is harmless. The cooling errors are correlated in time (Durbin–Watson 0.02), so the standard errors are slightly too small.

**Table 3.** Cubic fit coefficients. All terms have $p < 0.001$.

| Coefficient | Heating | Cooling | Global | Global std. err. |
|---|---|---|---|---|
| $c_0$ (intercept) | −18.792 | −7.364 | −17.559 | 0.076 |
| $c_1$ (ADC) | 0.2277 | 0.1495 | 0.2134 | 0.001 |
| $c_2$ (ADC²) | $-2.848\times10^{-4}$ | $-1.240\times10^{-4}$ | $-2.495\times10^{-4}$ | $1.34\times10^{-6}$ |
| $c_3$ (ADC³) | $2.107\times10^{-7}$ | $1.107\times10^{-7}$ | $1.869\times10^{-7}$ | $9.36\times10^{-10}$ |
| $N$ / AIC | 1078 / 773.8 | 4243 / 6914 | 5321 / 8557 | — |

**Table 4.** Residual checks for the cubic fits.

| Statistic | Heating | Cooling | Global |
|---|---|---|---|
| $F$-statistic | $4.06\times10^{6}$ | $1.38\times10^{6}$ | $3.21\times10^{6}$ |
| Durbin–Watson | 0.156 | 0.022 | 0.031 |
| Jarque–Bera | 7.80 | 325.3 | 468.2 |
| Skew / kurtosis | 0.21 / 2.98 | −0.53 / 2.16 | −0.72 / 2.80 |
| Condition number | $2.6\times10^{9}$ | $1.35\times10^{10}$ | $2.4\times10^{9}$ |

The heating residuals are close to normal (skew 0.2, kurtosis 3). The cooling and global residuals are slightly skewed, which means a small shape error remains, so the 95 % band is a rough guide.

## Heating vs. Cooling Hysteresis

We compared the two cubic fits over the shared range (ADC 309–849). The biggest gap is **1.15 °C** (1.15 % of full scale) and the average gap is **0.60 °C** (Figure 1). There are three likely causes:

1. **Thermal lag.** The NTC and the reference probe have different time constants $\tau = R_{\text{th}}C_{\text{th}}$, so one sensor trails the other while the temperature changes.
2. **Maths effect.** $\Delta T$ is the difference of two cubics, so it is a smooth wave crossing zero (near ADC ≈ 500).
3. **Water bath.** Without good stirring, the two probes see different temperatures.

The average gap is smaller than the reference uncertainty, so we do not claim real sensor hysteresis.

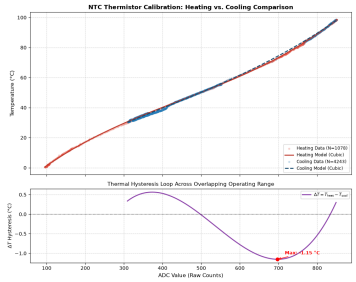



**Figure 1.** Heating vs. cooling fits and the gap $\Delta T$.

## Global Calibration and 95 % Prediction Band

We joined the heating data and the cleaned cooling data ($N = 5321$) and fitted one cubic, where $a$ is the ADC count:

$$T = 1.869 \times 10^{-7}a^3 - 2.495 \times 10^{-4}a^2 + 0.2134a - 17.559 \quad (1)$$

It has $R^2 = 0.999$ and RMSE 0.54 °C. The mean 95 % prediction half-width is **±1.06 °C** (Figure 2).

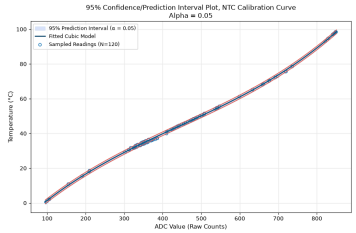



**Figure 2.** Global cubic fit with its 95 % prediction band.

## Theoretical NTC β-Model

The divider output is $V_{\text{out}} = V_{\text{cc}}R_{\text{f}}/(R_{\text{NTC}} + R_{\text{f}})$. The ADC uses $V_{\text{cc}}$ as its reference, so the reading does not change if the supply drifts:

$$\mathrm{ADC} = \mathrm{ADC}_{\max} \cdot \frac{R_f}{R_{\mathrm{NTC}} + R_f} \quad \Rightarrow \quad r \equiv \frac{R_{\mathrm{NTC}}}{R_f} = \frac{1023 - \mathrm{ADC}}{\mathrm{ADC}} \tag{2}$$

In an NTC, heat lifts charge carriers across the band gap $E_g$. This gives the Arrhenius law $R(T) = R_0 \exp[\beta(1/T - 1/T_0)]$, with $\beta = E_g/2k_B$ and $T_0 = 298.15$ K. Dividing by $R_f$ and taking the log gives a straight line:

$$\ln r = \frac{\beta}{T} + c, \quad c = \ln r_0 - \frac{\beta}{T_0}. \tag{3}$$

Let $A = 1/\beta$ and $B = -c/\beta$. Then solving for temperature gives the formula used in the logger:

$$T = \frac{1}{A \ln((1023 - \mathrm{ADC})/\ \mathrm{ADC}) + B} - 273.15. \tag{4}$$

## β-Model Validation

We fitted a straight line to $\ln r$ against $1/T$ (Figure 3). It gives $\beta = 3965.3$ K, $R^2 = 0.9993$ and an RMSE of **0.598 °C**, using only two parameters ($\beta$ and $r_0$).

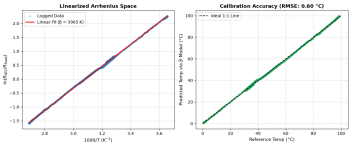



**Figure 3.** Straight-line Arrhenius fit (left) and β-model accuracy (right).

## Response Time of the Calibrated NTC

We dipped the probe repeatedly into ice water (0 °C) and room-temperature water, giving a square-wave input. After removing one spike (at 496 s) and the first cycle, 3 full periods remained. We converted ADC to °C with both the cubic and the β-model, then fitted Newton's law with a start delay to the 3 periods together:

$$T(t) = T_f + (T_i - T_f)e^{-(t-t_0)/\tau}. \tag{5}$$

The 3 periods share $\tau$, $T_i$ and $T_f$; each has its own delay $t_0$. The error comes from leaving one period out at a time (Figure 4).

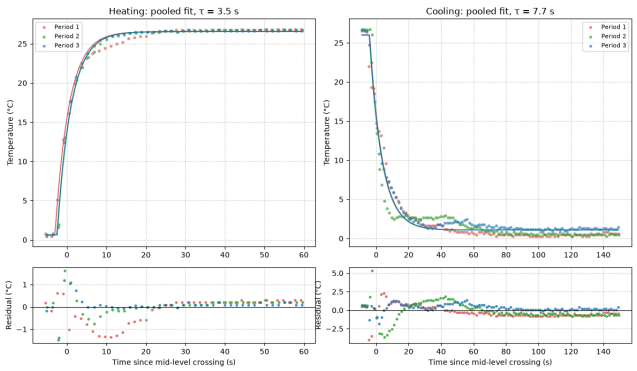



**Figure 4.** Pooled fit (cubic calibration) of the 3 heating and 3 cooling steps.

**Table 5.** Response time from the pooled fits (± is the leave-one-out error).

| Step | Calibration | $\tau$ (s) | 10–90 % (s) | RMSE (°C) |
|---|---|---|---|---|
| Heating | Cubic | 3.5 ± 0.3 | 7.8 | 0.47 |
| Heating | β-model | 3.3 ± 0.3 | 7.3 | 0.50 |
| Cooling | Cubic | 7.7 ± 2.2 | 16.8 | 0.92 |
| Cooling | β-model | 8.6 ± 2.2 | 18.9 | 1.02 |

The thermistor reaches 63 % of a step in about **3.4 s** when heating and about **8 s** when cooling, so it needs roughly 17 s and 40 s to settle (5 $\tau$). The two calibrations agree within 1 s, less than the error, so the choice does not matter. Cooling is slower and less certain: one period was much slower, and the ice bath warmed by about 2 °C during period 2, which a single exponential cannot follow (RMSE about 1 °C).

## Sensitivity and Error Theory

The sensitivity $dT/d\mathrm{ADC}$ is 0.102–0.195 °C/count for the cubic and 0.101–0.245 °C/count for the β-model (Figure 5). The sensor reads best mid-range and worse near both ends, where the divider output flattens.

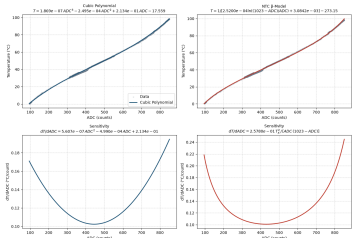


**Figure 5.** Fitted curves and sensitivity $dT/d\mathrm{ADC}$ for both models.

RMSE only measures scatter around the fit. To predict the error from its sources, we propagate three independent ones, $u_T^2 = \sum_i (\partial T/\partial x_i)^2 u_{x_i}^2$:

$$u_T = \sqrt{u_{\mathrm{ref}}^2 + (dT/d\mathrm{ADC})^2 u_{\mathrm{ADC}}^2 + \boldsymbol{g}^T \boldsymbol{C} \boldsymbol{g}}. \tag{6}$$

$u_{\mathrm{ref}} = 0.5$ °C is the DS18B20 accuracy. The ADC rounds to the nearest count ($u_{\mathrm{ADC}} = 0.5$), giving $u_{\mathrm{res}} = |dT/d\mathrm{ADC}| \cdot 0.5$, so a flat region reads coarsely. The fit term uses the parameter covariance $\boldsymbol{C}$ and $\boldsymbol{g} = \partial T/\partial$ params (with cross terms, as the coefficients are correlated).

**Table 6.** Error theory compared with the measured errors.

| Quantity (°C) | Cubic | β-model |
|---|---|---|
| Resolution, $\lvert dT/d\mathrm{ADC}\rvert_{\max}$ (°C/count) | 0.195 | 0.245 |
| $u_{\mathrm{res}}$ (max) | 0.097 | 0.122 |
| $u_{\mathrm{fit}}$ (mean / max) | 0.016 / 0.037 | 0.012 / 0.026 |
| $u_{\mathrm{ref}}$ | 0.5 | 0.5 |
| **$u_T$ (mean / max)** | **0.505 / 0.510** | **0.505 / 0.515** |
| Measured RMSE | 0.54 | 0.60 |
| Largest residual | 1.56 | 1.46 |

The reference error dominates; the fit and resolution terms add less than 0.1 °C each. The theory gives $u_T \approx 0.51$ °C, close to the measured RMSE (0.54 and 0.60 °C). The small extra part ($\sim 0.2$ –0.3 °C) is what the theory leaves out: thermal lag, hysteresis and model shape error. We quote the **largest** $u_T$ over the range as the standard error: **±0.51 °C (cubic)** and **±0.52 °C (β-model)**, or about ±1.0 °C at 95 % (two standard deviations), which agrees with the ±1.06 °C prediction band.

## Conclusion

Both models calibrate the thermistor to within the ±1 °C uncertainty of the DS18B20. The global cubic (4 parameters) has RMSE 0.54 °C and ±1.06 °C (95 % PI); the β-model (2 parameters, $\beta = 3965$ K, $R^2 = 0.9993$) has RMSE 0.60 °C. Error theory gives the same ±0.5 °C for both because the reference dominates. The cubic is slightly better inside 0–98 °C. The β-model is almost as good with half the parameters, works well outside the range, and gives a real physical constant.

**Limits.** The DS18B20 is only accurate to ±0.5 °C, so smaller errors cannot be proved. Lag between the two sensors adds error during rapid changes. Only three runs were usable (0–98 °C), so do not use the cubic outside this range.

**Recommendation.** Use the β-model as the main calibration, and the cubic only when you need the lowest error inside 0–98 °C. The heating/cooling gap (about 0.6 °C) is below the reference uncertainty, so one calibration works for both directions.